Intégrale n° 67 — Résolution

Énoncé. Calculer $I = \displaystyle\int_0^{\pi/2} \frac{\sin^2 x}{\sin x + \cos x}\,\mathrm{d}x$.

Notons $J = \displaystyle\int_0^{\pi/2} \frac{\cos^2 x}{\sin x + \cos x}\,\mathrm{d}x$. Le changement de variable $x \mapsto \frac{\pi}{2} - x$ donne $J = I$, et

$$2I = I + J = \int_0^{\pi/2} \frac{\sin^2 x + \cos^2 x}{\sin x + \cos x}\,\mathrm{d}x = \int_0^{\pi/2} \frac{\mathrm{d}x}{\sqrt{2}\,\sin\left(x + \frac{\pi}{4}\right)},$$

car $\sin x + \cos x = \sqrt{2}\sin\left(x + \frac{\pi}{4}\right)$. Une primitive de $\frac{1}{\sin u}$ est $\ln\tan\frac{u}{2}$ (sur $]0, \pi[$), donc

$$2I = \frac{1}{\sqrt{2}}\Big[\ln\tan\Big(\frac{x}{2} + \frac{\pi}{8}\Big)\Big]_0^{\pi/2} = \frac{1}{\sqrt{2}}\Big(\ln\tan\frac{3\pi}{8} - \ln\tan\frac{\pi}{8}\Big).$$

Or $\tan\frac{3\pi}{8} = 1 + \sqrt{2}$ et $\tan\frac{\pi}{8} = \sqrt{2} - 1 = \frac{1}{1+\sqrt{2}}$, donc la parenthèse vaut $2\ln(1+\sqrt{2})$ et $2I = \sqrt{2}\ln(1+\sqrt{2})$.

$$\boxed{I = \frac{\sqrt{2}}{2}\ln\left(1 + \sqrt{2}\right)}$$